

TABLE III
Comparison of Various Cleaner Systems

Cleaner system	Advantages	Important notes
Organic solvent cleaner	➤ Different types of impurities can be removed ➤ Wide process window	➤ High VOC value ➤ Flammability/flash point
Tenside cleaner	➤ Non-flammable ➤ Low VOC values	➤ Short bath lives ➤ Limits as to bath loading ➤ Large quantities of cleaner to be disposed of ➤ Critical no-clean cleaning ➤ Problems with subsequent coating processes may arise
MPC cleaner	➤ Non-flammable ➤ Low VOC content ➤ Different types of organic and inorganic impurities can be removed ➤ Impurities are insoluble ➤ Long bath lives ➤ No residue after drying	➤ Choose suitable mechanical support for cleaner (spray, ultrasonic, pressure liquid circulation)

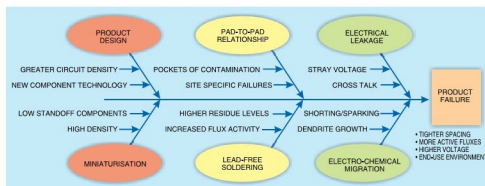


Fig. 1: Ishikawa diagram on the cause-and-effect relationship of high-end assemblies

solvent cleaner, tenside cleaner and MPC cleaner.

A comparison of these cleaner systems is listed in Table III.

Precleaning of assembled printed circuit boards

The increasing trend towards further miniaturisation of electronic assemblies along with smaller potential distances increases the requirement of cleaning the assemblies in case of lead-free solder processes. On the other hand, it also makes cleaning more difficult, given the reduced spacing between the components and circuit carrier. The Ishikawa diagram in Fig. 1 shows the various factors in detail which may cause failure in this context.

The subsequent protective coating should be applied as soon as possible after the cleaning process in order to contain the loading by further contamination. Upon a visual inspection with the naked eye, clean assemblies should

be free of residues and contamination, i.e., dirt, fluff, solder splashes, solder dross, etc. In this respect, the IPC-A-610D 'Acceptability of Electronic Assemblies' states the notes shown in Table IV in its foreword.

Assembly layout/printed circuit board layout

When coating an assembly, certain more or less defined areas have to remain uncoated besides the actual plug-in contacts, for diverse reasons. The causes may be:

1. Plug connectors, connector strips
2. Non-coatable components, such as relays and mechanical components
3. Areas to be bonded at a later stage
4. Areas or connectors for later contacts
5. Milling edges, etc

Depending on the positioning, these factors may be decisive, for example, for the later coating method used.

TABLE IV
Notes for Using Optical Magnification

Cleaner system	Important notes
Cleanliness (cleaning processes)	Magnification not required, see Note 1
Cleanliness ('no-clean' processes)	Note 1
Conformal coating/encapsulation	Notes 1, 2
Others (component and wire damage, etc)	Note 1

Note 1: Visual inspection may require magnification, for example, in case of closely spaced components or high-density assemblies, magnification may be needed to find out whether contamination affects shape, fit or function.

Note 2: If magnification is used it should be limited to 4X maximum.



Fig. 2: Plug saturated with coating material due to capillary action

In order to create a coatable layout, one should define any specific spacing and/or locations of components or plugs at this stage, in view of the coating process to be carried out at a later stage. Any failure at this stage could make it difficult to choose a cost-effective coating process or to achieve a coating in accordance with the specifications. The positioning of large components in closely adjacent areas defined to be free of coating should be avoided, if possible, as are accumulations of different component sizes. One should bear in mind that gaps attract liquid coating material via their so-called capillary action. While this is quite a desirable effect below components, it reverses into the negative in the case of plugs, as can be seen in Fig. 2. The plug has become saturated due to capillary action here despite a selective coating process,